

International AS and A-level

Chemistry (9620)

CH02 Organic 1 and Physical 1

Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL AS CHEMISTRY (9620) CH02 JUNE 2024

Students generally answered the questions well and are good at recalling test tube reagents and calculating equilibrium moles in reactions.

Some points to note to improve student outcomes in future examinations.

- Care must be taken when drawing structures or mechanisms. All relevant bonds need to be shown and curly arrows must start from a lone pair of electrons or a bond in mechanisms. Dipoles are not essential on mechanisms and can lead to marks being lost when they are incorrect.
- Students should not round data too much until the calculation is complete.
- Definitions, for example, types of isomers or formula, need to be learnt correctly.
- The origin of forces between organic molecules must be understood and explained.
- Writing balanced equations, especially for alcohols, needs to be developed in students' study of Chemistry.

QUESTION 01

- 1.1 This was answered well with the correct catalyst being selected. A common error was writing C_6H_{12} instead of $2C_3H_6$ as a product.
- 1.2 Over half of all students scored this mark, the common error was lack of discussion of products in the answer.
- 1.3 Just under half of all students lost this mark by not being precise enough about the bond that is broken. It is not enough to state that ultra-violet light makes free radicals or that it provides energy for the reaction.
- 1.4 Although the principle of free-radical substitution was well known, the main error was using molecular formula and removing the wrong hydrogen atom. It is not possible to show 2-chlorobutane when using molecular formula and many students lost the second mark because of this.
- 1.5 Only the best students scored this mark as they had drawn the correct formula for the radical formation of 2-chlorobutane. The mistake was joining two molecular formula C_4H_9 radicals.

QUESTION 02

- 2.1 Students found this mechanism harder than expected with only the best scoring full marks. Drawing of arrows must be clear and if dipoles are shown they must be correct. An error was to draw an arrow going from N to H instead of from the bond between H and N to the N^+ .
- 2.2 This was answered well. The main error was not including ammonia as a reactant in the calculation.
- 2.3 This was answered well with over two thirds of students scoring full marks. Rounding answers too soon was the main source of error.
- 2.4 This was answered well.

QUESTION 03

- 3.1 This was answered well with over 85% of students scoring full marks. The main error was getting the final ratio wrong after dividing by A_r .
- 3.2 This was answered well. The main error was writing CH_3 instead of showing all bonds or putting too many H atoms per carbon.
- 3.3 Two thirds of students scored this mark. The main error was drawing propene instead of cyclopropane or giving a structure with an incorrect name.
- 3.4 This definition was only correctly stated by half of students. The main error was stating molecular formula instead of structural formula.
- 3.5 Students found this question hard with only the best students scoring full marks. They need to be very clear that both C of $\text{C}=\text{C}$ have two different groups attached. Many more students scored the mark about restricted rotation of $\text{C}=\text{C}$.
- 3.6 Even though the principles of CIP rules were known, the explanations were incorrect, and this was a challenging question for students. The main error was to state molecular mass or atomic mass instead of atomic number, and not all students compared groups on each carbon of $\text{C}=\text{C}$. Not all students were clear that the Z-isomer needs to have the priority groups on the carbon atoms at both ends of the double bond on the same side of $\text{C}=\text{C}$.

QUESTION 04

- 4.1 This mechanism was known better than others in the paper with nearly half of all students scoring 5 out of 6 marks for the mechanism and colour change observed in the reaction. All arrows must start from a lone pair of electrons or bond in the mechanism. Not as many students could explain that the colour change happened because the bromine molecule was removed from the solution during the reaction, many students just restated that it was an addition reaction.
- 4.2 This was a challenging question for students, with approximately 20% of students making good attempts at naming the structure. The diene structure was answered better for the second mark, but students struggled with drawing the skeletal formula correctly.

QUESTION 05

- 5.1 Approximately half of all students were able to draw the displayed formula correctly. The main error was not drawing a bond between O and H or missing the $\text{C}=\text{C}$ that is needed to form addition polymers.
- 5.2 This was not answered well with students discussing intermolecular forces instead of non-polar or unreactive $\text{C}-\text{C}$ bonds.
- 5.3 The use of plasticisers and explanations of how plasticisers work was answered better by students than Question 05.2

QUESTION 06

- 6.1 This was answered well by a third of students who scored full marks. The common error by other students was missing out heating under reflux for the condition of the reaction. Marks were also lost by poor drawing of structures.
- 6.2 This was known well with many students scoring the mark. A surprising number did not calculate M_r correctly.
- 6.3 This was not answered as well as expected. A common error was doing O–H–C instead of O–H–O for the hydrogen bond. Not all lone pairs and dipoles were shown, or the hydrogen bond was not a straight line with the O–H bond.
- 6.4 Approximately two thirds of students were able to put the correct formulas in the equation. The common error was balancing using 5O_2 instead of $4\frac{1}{2}\text{O}_2$
- 6.5 This was answered well.

QUESTION 07

- 7.1 This was answered well, and chemical tests are known by students. Marks were lost by stating no observation which implies that the reaction has not been looked at instead of no visible observation or no visible change.
- 7.2 Not as well-known as Question 07.1, marks were lost by stating gas formed instead of the observation of bubbles or fizzing.
- 7.3 Just under half of all students scored this mark. The main error was not quoting the correct range from the Data sheet and that it represented the O–H bond in carboxylic acids.
- 7.4 This was unexpectedly difficult for students with only the best students knowing how to use the fingerprint region to confirm identity of compound.

QUESTION 08

- 8.1 This was answered well, but some students lost marks by not stating rates of reactions are the same and that concentrations are equal.
- 8.2 Nearly 80% of students scored these marks showing that calculating equilibrium moles is known well.
- 8.3 A surprising number of students discussed rate instead of equilibrium yield. Some stated the reaction shifting instead of equilibrium.
- 8.4 This was answered well with most students scoring full marks. Common mistakes were missing out zeros in the calculation or not inputting data in calculator correctly. Pleasingly very few round brackets were seen but the K_c expression must be written with compound formula in square brackets.
- 8.5 About half of all students were able to correctly suggest that activation energy was high, or temperature was too low.
- 8.6 Approximately two thirds of students scored the mark about lower pressure. But only a third were able to successfully explain in terms of the frequency of successful collisions. Some students discussed shifting equilibrium position, but the question was about time and needed an explanation based on the rate of reaction.
- 8.7 This was answered well.